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EXERCISE

# Lecture 1: Introduction

AI508 — Algorithmic Game Theory

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Fall - 2026



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# 1 What is Game Theory

## Definition: Game Theory (very brief)

Created by John Nash, Game Theory is the study of decision making in settings with multiple rational agents whose outcomes depend on each other's choices.

## Definition: Algorithmic Game Theory (also very brief)

Studying Game Theory with a computational lense

# 2 System/Mechanism Design

Design a good system such that it cannot be exploited easily (2012 olympics badminton china vs korea)

# 3 Cooperations and Incentives

Given something like the The Kyoto Protocol, where all countries should reduce the global C<sub>O</sub>2 emission. Could it be beneficial for some countries to just keep emitting and let others reduce, given the cost and how much others reduce.

# 4 Image Generation with GAN's

$$\min_{\theta} \max_{\phi} (G_{\theta}, D_{\phi}) = \mathbb{E}_{x \in p_{\text{data}}} [\log D_{\phi}(X)] + \mathbb{E}_{z \in p(z)} [\log(1 - D_{\phi}(G_{\theta}(z)))]$$

# 5 What we will cover

- Auctions
- Selfish routing
- Bimatrix games and equilibria
- Decentralized learning pf equilibria
- Introduction to minmax optimization and GAN's
- Cooperative games and Shapley values (week 46)
- Extensive form games and analyzing (Kuhn) Poker

# 6 Learning outcomes

- Mathematically formulate any (relevant) problem as a multivalent learning problem
- Analyze effectiveness of systems/mechanisms
- In real-world strategic scenarios, identify the players and the game
- Identify the information available and use a relevant algorithm
- Identify

## 7 Relevant Material

**Game Theory Basics:** Lecture Notes on **Algorithmic Game Theory** by Tim Roughgarden, **Algorithm Design and Analysis** by Noam Nisan, Tim Roughgarden, Éva Tardos.

**Misc:** Relevant papers/readings will be posted for certain weeks

For further information/exercises: CS364A: Algorithmic Game Theory

## 8 Exam

29th of january